Multi-Agent Web Accessibility Improvement System

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ABSTRACT

Despite the growing demand for accessible web experiences, current tools like Axe and Lighthouse remain limited to static, rule-based diagnostics without contextual reasoning or user-centered remediation. We present a multi-agent, LLM-powered framework for automated web accessibility auditing and fix generation, simulating diverse user personas (e.g., visually impaired, motor impaired, color blind) to surface nuanced, real-world usability challenges. Our architecture decomposes the task across specialized agents—semantic, contrast, image captioning—whose suggestions are aggregated and explained through a central Fixing Agent.

A key contribution of this work is the construction of a high-quality, explainable fix dataset comprising natural language remediations, each tied to a specific WCAG violation, UI snippet, and DOM element. To ensure real-world applicability, we integrated the system into an interactive feedback platform, collecting 548 human judgments across 11 websites and 14 violation types. The resulting dataset not only demonstrates high user agreement (84.07%) but also serves as a reusable benchmark for training and evaluating next-generation accessibility systems. By bridging diagnosis, explanation,

and feedback collection, our system offers a scalable foundation for reproducible and participatory accessibility research.

KEYWORDS

web accessibility, multi-agent systems, large language models, WCAG compliance, human-AI collaboration, inclusive design, accessibility benchmarking

INTRODUCTION

Problem Statement

Ensuring web accessibility is challenging in modern UI/UX design. Although existing tools like Google Lighthouse and Axe DevTools provide automated, rule-based checks for compliance with the Web Content Accessibility Guidelines (WCAG), they offer limited contextual understanding and fail to simulate real user experiences. More importantly, these tools only diagnose accessibility issues, and lack AI-driven capabilities to automatically identify and provide actionable guidance on how to resolve them. As a result, there exists a significant gap between identifying accessibility violations and guiding for implementing effective fixes.

To address this gap, our project focuses on building a multi-agent system that not only identifies accessibility violations but also recommends intelligent, context-aware fixes.

Motivation

Web accessibility is not a feature—it's a necessity. With over one billion people worldwide living with some form of disability, ensuring equitable access to digital platforms is a moral imperative. Yet, many websites and applications continue to remain hard to use and navigate for individuals with visual, auditory, or cognitive impairments.

While tools like Axe and Google Lighthouse provide rule-based accessibility testing, they lack the contextual sensitivity required to understand real user experiences. As a result, they often fail to capture the nuanced, personalized challenges users face. Moreover, they do not offer recommendations to help developers resolve the identified issues.

This gap presents an opportunity for human-AI collaboration. By incorporating AI, we can simulate diverse user perspectives. Intelligent agents can assist designers and developers not only in detecting accessibility issues but also by recommending actionable, user-centered fixes. Through this, we aim to empower teams to build more inclusive digital experiences with greater efficiency and empathy.

LITERATURE SURVEY & RELATED WORK

Digital-accessibility research has advanced through three main waves: rule-based auditing, data-driven UI understanding, and, most recently, LLM-powered repair. Our project positions itself at the intersection of the latter two.

Rule-based Auditing (2004–present)

Early tools such as WAVE, AXE, and Lighthouse translated WCAG heuristics into deterministic checks. Although these tools are fast and provide clear failure modes, their evaluative coverage is limited. They struggle with semantic issues (e.g., misleading link text) and compound faults (e.g., colour-contrast *plus* focus-order errors) and cannot propose meaningful fixes.

Dataset-driven UI Understanding

The release of large annotated UI datasets such as **WebUI** and **Rico** enabled computer-vision and graph-based methods to reason over screenshots, DOM trees, and interaction traces. Researchers exploited these corpora to predict component semantics, identify design patterns, and detect anti-patterns beyond traditional static rules.

ML and LLM-based Evaluation and Repair

Recent work harnesses deep learning and foundation models for both detection and remediation:

- **Vemuri** (2024) showed that GPT-style models can rewrite HTML to remove violations such as contrast issues and missing landmarks.
- **Yu *et al.*** (2025) restructured web content in real time to improve navigation for screen-reader users.
- **ScreenAudit** (2025) extended mobile auditing by prompting LLMs to diagnose TalkBack-specific issues and generate developer-friendly rationales.
- The **W3C AI & Accessibility Symposium** (2023) highlighted the need for persona simulation and multimodal analysis.
- **Roselli** (2024) cautioned against fully automated overlays that may degrade UX and create legal-compliance risks.

Human–AI Collaboration and Participatory Evaluation

Studies by Hämäläinen *et al.* (2023) and Lee *et al.* (2022) demonstrated that human–AI co-creation improves model fidelity and trust. We adopt a similar strategy by evaluating 548 user votes across 14 violation categories, achieving 84.07% inter-reviewer agreement.

Our Contribution

Existing LLM-based systems are often monolithic and rarely publish curated fix corpora. Our multi-agent system:

- assigns *specialist* agents (semantic, contrast, image-captioning) to distinct violation types,
- aggregates their outputs through a central *Fixing Agent*, and
- releases a *fix dataset* for reproducible training and evaluation.

HCI-DRIVEN CONTRIBUTIONS AND RESEARCH QUESTIONS

We propose a multi-agent, large language model (LLM)-driven framework for enhancing web accessibility through explainable, actionable, and user-centered recommendations. Unlike conventional rule-based tools such as Axe and Lighthouse that merely diagnose WCAG violations, our approach simulates real-world accessibility challenges by representing diverse user personas (e.g., color-blind or motor-impaired users) and offering targeted, intelligible fixes. This work lies at the intersection of Human-Computer Interaction (HCI) and Machine Learning (ML), combining model-driven inference with participatory evaluation to improve the accessibility of user interfaces in a grounded, inclusive, and transparent manner.

The first research question we addressed is: *Can we construct an LLM-augmented dataset of accessibility fixes using web UI data?* To answer this, we began with WebUI7k, a dataset of web UI screenshots and corresponding HTML code, and enriched it with accessibility annotations using Axe DevTools. These annotations provided a rule-based foundation for detecting WCAG violations. We then fine-tuned and prompted a suite of LLM-based agents—specialized for semantics, contrast, and image captioning—to generate accessibility fixes contextualized to each violation. The fixes were expressed in natural language and targeted specific DOM elements, enabling not just automated correction but also explainability for developers.

The resulting dataset pairs each web snippet with the original violation, a natural language fix, and metadata capturing both the agent's rationale and the UI context. This new corpus bridges a critical gap in existing accessibility research: datasets that not only flag compliance issues but also demonstrate how to fix them with human-like reasoning. It enables reproducible evaluation of future models and serves as a foundation for training accessibility-aware generative systems grounded in real-world design patterns.

The second research question explored whether multi-agent LLM systems can produce meaningful and user-aligned accessibility recommendations. We designed a modular architecture where each agent specializes in identifying and resolving a particular class of accessibility concern. For example, the Contrast Agent detects color violations and suggests improved schemes that meet WCAG contrast thresholds, while the Semantic Agent targets missing or misused ARIA attributes, and the Image

Captioning Agent proposes descriptive alternative text. In parallel, Simulation Agents model different disability perspectives (e.g., visual, motor), identifying interaction issues that may not surface in rule-based scans.

These agents operate independently, and their outputs are then aggregated by a Fixing Agent into a unified, readable accessibility report per website. This architecture allows our system to simulate diverse user needs and produce rich, context-sensitive design suggestions. The explainable, element-specific outputs foster developer trust and usability, aligning well with HCI principles of interpretability and transparency in intelligent systems. Moreover, the system demonstrates how AI can serve as an assistive partner in participatory design rather than a replacement for human insight.

The third research question asks: *Do users perceive AI-generated fixes as actionable and useful in practice?* To evaluate the effectiveness of our recommendations, we conducted a user study using a custom-built web platform. Participants were shown real UI snippets containing accessibility issues, followed by agent-generated fixes and explanations. Each fix was rated along three dimensions: usefulness, clarity, and actionability. The study involved over 140 reviewers and spanned 11 websites with 14 categories of violations, resulting in 548 individual fix assessments.

The evaluation revealed a high degree of user alignment with our system's suggestions. The average agreement rate across all fixes was 84.07%, with certain categories such as missing landmarks and missing alt text for images achieving over 90% agreement rate. These findings validate the system's ability to produce not just technically correct but also human-usable recommendations. Importantly, the evaluation surfaced remaining challenges—such as heading hierarchy or tabindex misconfigurations—where fixes require deeper semantic or contextual understanding. These insights highlight the value of hybrid AI-human collaboration.

In summary, our contributions include: (1) a new LLM-augmented dataset of annotated accessibility violations and human-readable fixes; (2) a multi-agent architecture that simulates diverse disability perspectives and produces contextual recommendations; and (3) a participatory evaluation pipeline that validates system output through real user feedback. This work advances the integration of ML into accessible design by centering on user experience, explainability, and actionable system output—core principles at the intersection of HCI and AI.

DESIGN AND METHODOLOGY OF MULTI-AGENT WEB ACCESSIBILITY IMPROVEMENT SYSTEM

Our methodology combines system design, dataset construction, and user-centered evaluation to assess the effectiveness system and usability of the resultant dataset. This approach consists of various training phases for each agent on augmented datasets and a user feedback system to evaluate the generated fixes.

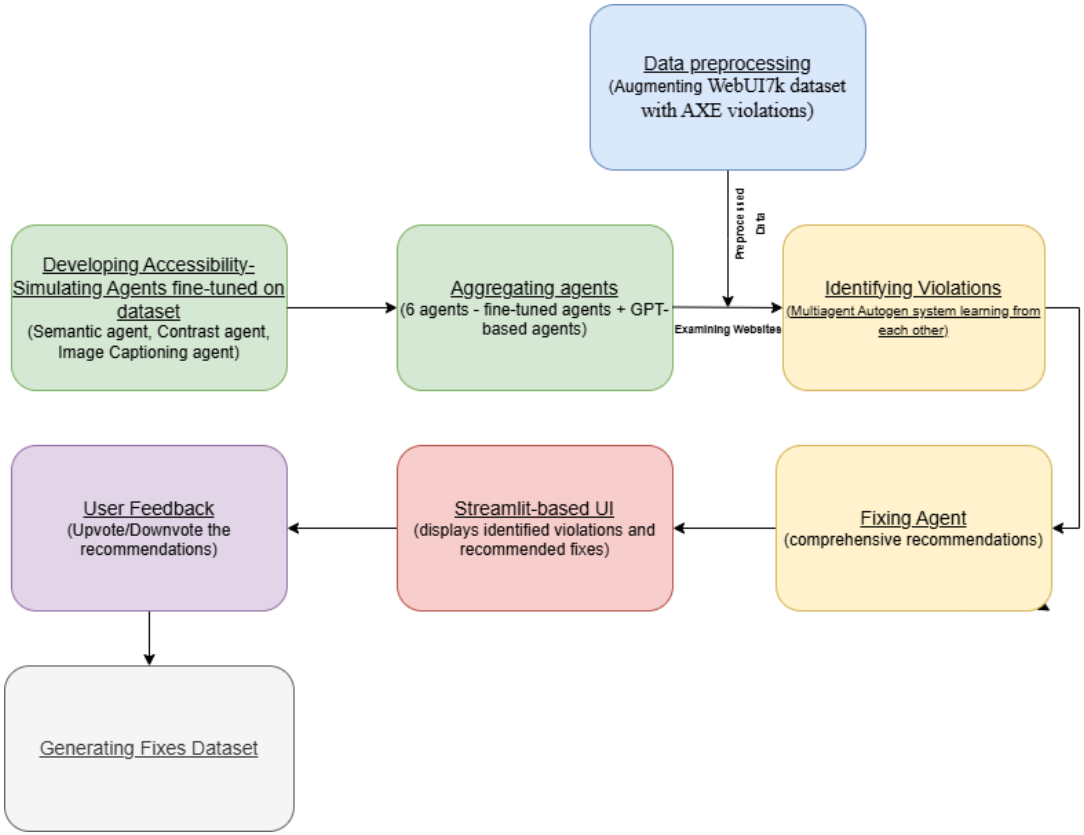

Figure 1: Workflow of multi-agent system to generate a fix dataset

Dataset Construction

We created a dataset by adding AI-generated accessibility fixes to the WebUI7k corpus to facilitate both agent development and user evaluation. First, we annotated web UI examples for accessibility violations using AXE DevTools. An initial layer of ground truth was provided by these annotations. After that, our agents produced natural language fix recommendations for every infraction, creating a hybrid dataset that blends LLM-driven remediations with rule-based detection. The HTML snippet,

a screenshot of the impacted user interface, metadata regarding the identified infraction, and a structured, human-readable fix created by one or more agents are all included in each sample in the dataset. This dataset, which spans 11 real-world websites and contains fixes for 14 WCAG violation types, was utilized as evaluation material and as input for training agents.

System Architecture

The modular, agent-based architecture of our system allows each agent to represent a distinct accessibility viewpoint, such as color blindness, low vision, or semantic navigation challenges. These agents, which can be prompted or fine-tuned using GPT-4, power large language models (LLMs), which work together to identify violations and produce recommendations for natural language fixes using code snippets. Key components of the system include:

(1) **Preprocessing and Annotation**: The WebUI7k dataset served as the basis for our analysis, with accessibility violations systematically detected using the AXE DevTools engine.
(2) **Multi-Agent Chat**: Built using the AutoGen framework to simulate deliberation among agents for improved fix quality. These agents were:
    - *Semantic Agent (T5-based)*: Identifies improper heading structures and semantic inconsistencies in content layout.
    - *Contrast Agent (T5-based)*: Detects color contrast violations that fall below WCAG thresholds.
    - *Image Captioning Agent (T5-based)*: Generates or verifies descriptive alt-text for images to ensure non-visual accessibility.
    - *Visually Impaired Agent (GPT-4-based)*: Simulates non-visual browsing patterns to detect missing ARIA landmarks, insufficient label associations, and non-descriptive navigation flows.
    - *Motor Impaired Agent (GPT-4-based)*: Evaluates keyboard navigability, tab order integrity, and interaction effort required by users with limited mobility.
    - *Color Blind Agent (GPT-4-based*): Assesses the UI for problematic color pairings that hinder accessibility for users with red-green, blue-yellow, or monochromatic vision deficiencies.
    - *Fixing Agent (GPT-4-based)*: Serves as the multi-agent architecture's main aggregator, combining the outputs of all specialized agents to suggest logical, situation-specific accessibility solutions. In addition to resolving conflicting recommendations and performing cross-agent reconciliation, the Fixing Agent creates final recommendations in natural language that are both actionable and compliant with WCAG guidelines. It serves as the agent loop's last decision-maker, guaranteeing that the composite fix maintains the integrity of the interface while taking into consideration various disability viewpoints.

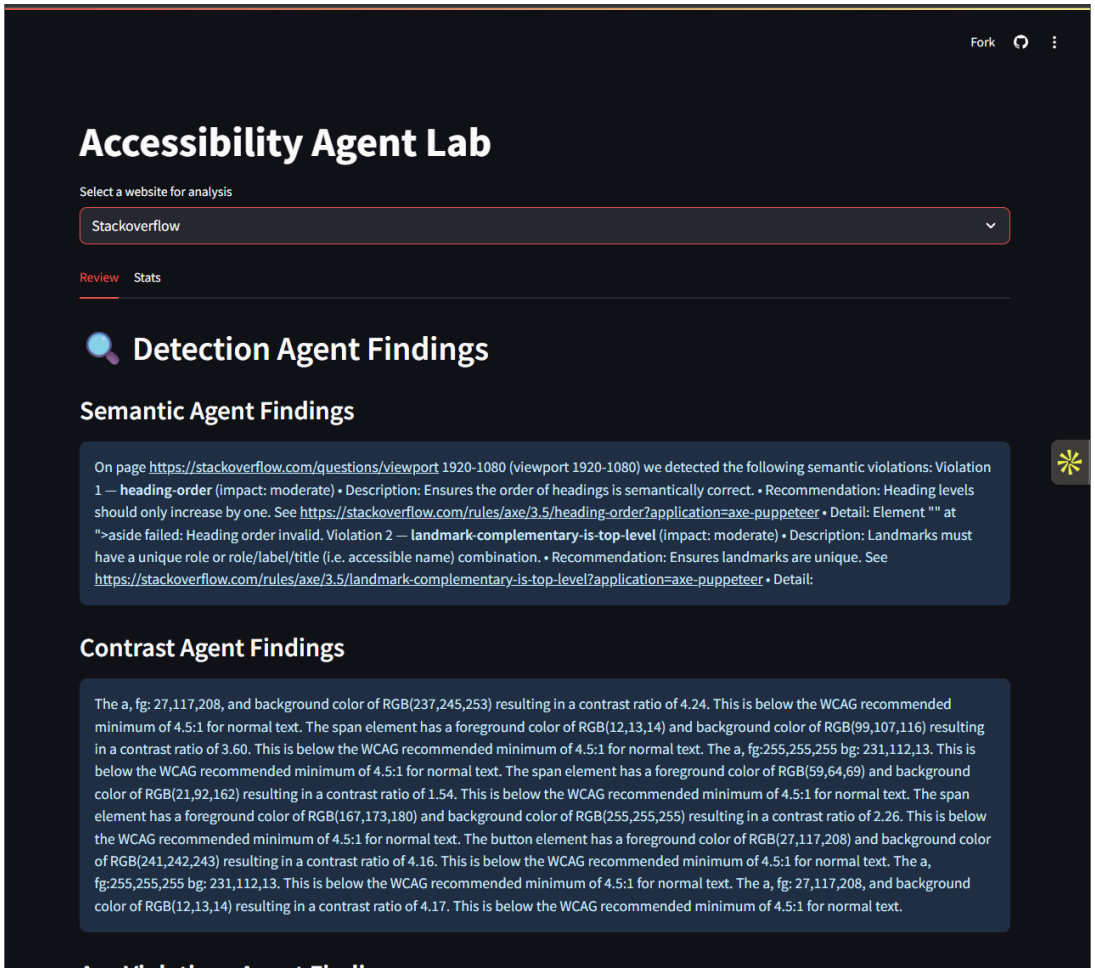

Figure 2: Web UI of user feedback system

(3) **Feedback Interface**: Using Streamlit, we created a simple web-based interface to facilitate user evaluation. Eleven real-world websites had pre-configured evaluation pages hosted by the

platform, each of which included a carefully selected list of accessibility issues found and the corresponding fixes created by the agents. After reviewing each fix in context, participants were asked to rate its appropriateness and effectiveness using a straightforward upvote/downvote mechanism. Users could see the total number of votes cast for that specific fix after responding, which promoted openness and a feeling of involvement. This configuration made it possible to collect structured, asynchronous feedback at scale, facilitating both qualitative understanding of user judgment and quantitative analysis of consensus.

Table 1: High-Performing Accessibility Fixes (>90% Agreement)

Violation Category	Agreement (%)
Document lacks a main landmark	100.00
All links should have discernible text	90.91
`<html>` element lacks a `<lang>` attribute	90.91
Inaccurate alt text for images	90.56
All links should have discernible style	90.00

[1]https://ai-assistant-for-accessibility.streamlit.app/

Table 2: Moderately-Performing Accessibility Fixes (80-90% Agreement)

Violation Category	Agreement (%)
Heading order should only increase by one	87.83
Missing alt text for images	85.90
Content outside landmarks	82.16
Color contrast minimum ratio not met	80.91
Landmarks must be unique	80.27
Page lacks a level-one heading	80.23
Elements with tabindex greater than 0	80.00

EVALUATION AND EXPERIMENTAL RESULTS

Evaluation Study Overview

Our evaluation study involved a feedback-driven accessibility review using a dedicated online platform hosted via Streamlit.[1] The objective was to test the robustness and effectiveness of our automated accessibility agents across diverse real-world websites. The evaluation spanned across 11 distinct websites, each evaluated on approximately four accessibility issues with their corresponding recommended fixes. The websites considered were the homepages of the following websites: Google, Wikipedia, Stanford, Microsoft, Reddit, Stackoverflow, GNU, Harvard, IETF, Netflix, and Quora. A total of 14 unique violation categories were tested. The evaluation gathered substantial community feedback from an estimated 142 individual reviewers (approximately 13 reviewers per webpage), yielding a cumulative 548 votes (upvotes and downvotes combined).

Overall, the community reviewers demonstrated a strong consensus, with an average agreement rate over all websites considered, of approximately 84.07% regarding the recommended fixes proposed by our automated agents. We categorize fixes into three distinct segments based on community feedback measured through upvotes, downvotes, and overall percentage agreement.

High-Performing Accessibility Fixes

Certain violation categories demonstrated exceptionally high levels of agreement from community reviewers, reflecting superior performance of our automated accessibility agents. These fixes received over 90% approval and minimal dissent, indicating robust trust and reliability in the automated suggestions (see Table 1). Notably, fixes related to clearly defined HTML structure and semantic correctness (such as missing main landmarks and discernible link texts) received unanimous or near-unanimous support, confirming that automation in these straightforward areas is highly effective.

Moderately-Performing Accessibility Fixes (80-90% Agreement)

A set of violation categories exhibited moderately high community consensus, typically scoring between 80% to 90% approval. These issues often involved slightly subjective or complex accessibility

improvements, which likely contributed to a more varied response from reviewers (Table 2). For instance, fixing the heading order and missing alt texts requires interpretative judgments, which might explain the lower but still positive agreement scores. These are areas where automated approaches are valuable, yet occasionally need further manual validation.

Accessibility Fixes Requiring Additional Human Review (<80% Agreement)

Some violation categories showed lower consensus (below 80%), suggesting these fixes might benefit substantially from human judgment or further refinement of automated logic. The disagreements generally arose from complex or nuanced interpretations of ARIA roles, proper usage of tabindex, or semantic importance, such as the necessity of a `<title>` element (Table 3). These discrepancies underscore the ongoing need for human oversight in ensuring web accessibility where automation alone might not fully capture user needs or context-specific appropriateness.

Table 3: Accessibility Fixes Requiring Further Human Review (<80% Agreement)

Violation Category	Agreement (%)
Inappropriate ARIA role attributes	78.34
Document lacks a non-empty `<title>`	63.64

These insights pinpoint areas where automated accessibility agents can reliably augment human judgment and identify gaps where human expertise remains critical. Future improvements to our system will particularly focus on addressing violation categories with lower reviewer consensus to enhance reliability and trustworthiness of the automated accessibility recommendations.

FUTURE WORK

Future developments will primarily focus on refining the effectiveness, explainability, and practical integration of our automated accessibility enhancement system.

Initially, we plan to enhance the accuracy of our accessibility agents through reinforcement learning techniques informed by validated human feedback. Each automated suggestion will be accompanied by clear, natural language explanations referencing the relevant Web Content Accessibility Guidelines (WCAG) and detailing specific changes through illustrative examples. Additionally, we will improve our feedback mechanisms by allowing practitioners to provide comprehensive textual feedback alongside simple agree/disagree interactions.

Subsequently, the scope of our agents will be broadened to encompass a more diverse array of accessibility needs, including cognitive considerations, auditory impairments, and sensitivity to motion. We also intend to embed real-time accessibility evaluations directly into common web browsers through a cross-platform extension, enabling immediate and contextualized feedback during the development phase. Additionally, facilitating URL-based scans will further streamline the review process, enhancing the overall user experience.

Further improvements will involve seamless integration of our accessibility checks within popular development environments such as VS Code and JetBrains IDEs, providing inline feedback and convenient remediation options directly in the development workflow. Moreover, expanding agent

training to include extensive mobile UI libraries will ensure consistency and reliability across different devices. Integrating accessibility audits within continuous integration (CI) pipelines via GitHub Actions will proactively prevent regressions and uphold accessibility standards across deployments.

Finally, we envision establishing a multimodal design environment that leverages generative AI, voice commands, and visual simulations to enable designers to experience web interfaces through the perspectives of various accessibility personas. This integrated studio, alongside the previously outlined enhancements, aims to provide a comprehensive, user-centered ecosystem that promotes accessibility awareness, automation, and continuous improvement throughout the entire development lifecycle.

REFERENCES

[1] Vemuri, V. P. K. (2024). *Advancing Web Accessibility: Leveraging Generative AI for Enhanced WCAG Compliance*. Journal of Artificial Intelligence & Cloud Computing. https://doi.org/10.47363/JAICC/2024(3)E261

[2] Yu, Y., et al. (2025). *From Cluttered to Clear: Improving Web Accessibility for Screen Reader Users in E-commerce with Generative AI*. arXiv:2502.18701. https://arxiv.org/abs/2502.18701

[3] Madu, I., & Nzenwa, C. (2024). *AI in Web Development: Enhancing Accessibility*. SSRN. https://doi.org/10.2139/ssrn.4949516

[4] Wu, J., et al. (2023). *WebUI: A Dataset for Enhancing Visual UI Understanding with Web Semantics*. arXiv:2301.13280. https://arxiv.org/abs/2301.13280

[5] Deka, B., et al. (2017). *Rico: A Mobile App Dataset for Building Data-Driven Design Applications*. In Proc. UIST 2017, 845–854. https://doi.org/10.1145/3126594.3126651

[6] Hämäläinen, H., et al. (2023). *Evaluating Large Language Models in Generating Synthetic HCI Research Data*. Proc. CHI 2023. https://doi.org/10.1145/3544548.3580688

[7] Lee, N., et al. (2022). *CoAuthor: Designing a Human-AI Collaborative Writing Dataset*. arXiv:2201.06796. https://arxiv.org/abs/2201.06796

[8] Xie, Y., et al. (2025). *ScreenAudit: Detecting Screen-Reader Accessibility Errors in Mobile Apps with LLMs*. Proc. CHI 2025.

[9] W3C WAI. (2023). *AI & Accessibility Research Symposium Report*. https://www.w3.org/WAI/about/projects/ai-accessibility/

[10] Roselli, A. (2024). *No, "AI" Will Not Fix Accessibility*. Blog article. https://axesslab.com/ai-will-not-fix-accessibility

[11] Farfan, C., et al. (2025). *Automated Evaluation of Accessibility Issues of Webpage Content*. Scientific Reports, 15, 92192.

[12] VKTR Research. (2025). *The Open-Source AI Accessibility Checker Holding LLMs Accountable*.